

Aliah University

Autumn Semester Examination - 2024

B.Tech 4th year, 7th semester Examination

Paper Name: Machine Learning and Soft Computing

Paper Code: CSEUGPC24

Full Marks: 80

Time: 3 hrs

Group-A

Answer any five

(2x5=10)

1. Define Binary Fuzzy relationship. (2) (CO1), (Rem)
2. Write the basic steps for a Genetic Algorithm problem. (2) (CO2), (Eva)
3. Compare Genotype and Phenotype with simple examples. (2) (CO3), (Und)
4. What is Gradient-Descent? (2) (CO4), (Rem)
5. Discuss the concept of a Self-organizing map. (2) (CO5), (Und)
6. Analyze uniform crossover and single point crossover in Genetic Algorithm. (2) (CO2), (Ana)

Group-B

Answer any four

(5x4=20)

7. Give a brief explanation for the conceptual working flow diagram of a fuzzy-based system. (5) (CO1), (Ana)
8. Maximize the function $f(x) = 3x^2 - 5x + 1$, where $x = 4, 9, 14, 17$ with (Chromosome size = 5 such that (i) selection operation (Rank Selection), (ii) Uniform crossover, (iii) Up to two iterations. (5) (CO3), (Eva)
9. Consider the following grading system for a course: (i) Excellent = Marks ≤ 90 , (ii) Very good = $75 \leq$ Marks ≤ 90 , (iii) Good = $60 \leq$ Marks ≤ 75 , (iv) Average = $50 \leq$ Marks ≤ 60 , (v) Poor = $35 \leq$ Marks ≤ 50 , (vi) Bad = Marks ≤ 3 . Establish the fuzzy Grading System. (5) (CO2), (App)
10. Consider the following two fuzzy sets A and B defined over a universe of discourse $[0,5]$ of real numbers with their membership function: $\mu_A(x) = \frac{x}{1+x}$ and $\mu_B(x) = 2^{-x}$. Determine the membership functions of the (i) $\overline{A}, \overline{B}$, and (ii) $A \cap B$ and draw them graphically. (5) (CO2), (Eva)

11. Write short notes on activation functions: (i) Threshold, (ii) Sigmoid, (iii) Hyperbolic Tangent.

(5) (CO4), (Ana)

Group-C

Answer any five

(10x5=50)

12. Describe the Roulette-wheel selection for the GA-based selection operator. Suppose, two relations A and B are given as follows: A: 'x is considerably larger than y', B: 'x is very close to y'. Find the fuzzy relationships between 'x is considerably larger than y' or 'x is very close to y'. Analyze the mamdani Fuzzy inference system? (2+4+4=10) (CO3, CO2), (Rem, Eva, Ana)

A	y1	y2	y3	y4
x1	0.6	0.7	0.4	0.6
x2	0.5	0.6	0.3	0.7
x3	0.2	0.1	0.3	0.8

B	z1	z2	z3
y1	0.5	0.9	0.3
y2	0.8	0.1	0.8
y3	0.9	0.5	0.2
y4	0.4	0.2	0.4

13. Draw a very clear 5-4-4 ANN architecture with explaining all its components. Give the parameters requirement for the good clustering algorithms. What are Core and Support in fuzzy logic?

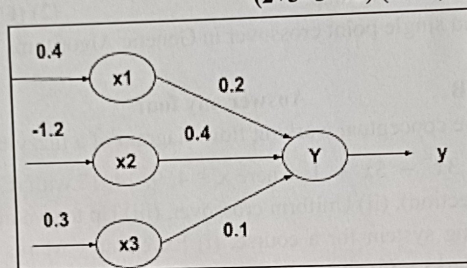
(5+3+2=10) (CO4, CO4, CO2), (Cre, Ana, Rem)

14. Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of 6 genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a * b) + (c * d) + (e * f) - (g * h)$. Let the initial population consist of four individuals with the following chromosomes:

$x1 = 7\ 2\ 4\ 1\ 3\ 5\ 3\ 2$; $x2 = 9\ 7\ 1\ 2\ 1\ 6\ 0\ 1$; $x3 = 5\ 3\ 2\ 2\ 1\ 2\ 8\ 5$; $x4 = 7\ 1\ 8\ 5\ 2\ 4\ 9\ 4$. Use the following (i) Evaluate the fitness of each individual, (ii) Cross the fittest two individuals using one-point crossover at the middle point, (iii) Evaluate the fitness of the new population with the best four chromosomes (two-old and two-new) (iv) Perform (ii) to (iii) up to three iterations. What is the difference between L1 regularization and L2 regularization?
(7+3=10) (CO3, CO4), (Eva, Rem)

15. What is the relationship between ROC and AUC? Give explanations about convex optimization and Linearity vs. Non-linearity concepts in ML. Discuss a covariance matrix with suitable examples. Write about bias and variance tradeoff.
(2+4+2+2=10) (CO4), (Rem, Und, Rem)

16. What is singular value decomposition (SVD)? How is it related to principal component analysis (PCA)? Find the value of outcome (y) from the given network using ReLU activation function.
(2+3+5=10) (CO4), (Rem, Und, Ana)



17. Analyze the Training and Testing operation of a Recurrent Neural Network. What are the steps involved in designing a fuzzy logic controller? Explain the different types of membership function used in the fuzzification process.
(3+3+4=10) (CO4, CO1, CO1), (Ana, Und, Ana)

Odd (Autumn) Semester Examination 2025
Paper Code: MBAUGHU01; Paper name: Industrial Economics and Management
VIIth Semester (EEN/CEN/MEN/CSE)
Full Marks: 80; Time: 3Hrs.

(The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions)

(1 x 10 = 10)

- I. If the quantity demanded of a commodity is unresponsive to change in prices, then the demand of that commodity is _____.
(a) Perfectly inelastic (b) Unit elastic (c) Elastic (d) Inelastic
- II. The following is (are) non-verbal communication.
(a) Facial Expression (b) Appearance (c) Posture (d) All of the above
- III. The term marketing refers to _____.
(a) New product concepts and improvements
(b) Advertising and promotion activities
(c) A philosophy that stresses customer value and satisfaction
(d) Planning sales campaigns
- IV. _____ is known as "the father of scientific management."
(a) Fredrick W. Taylor (b) Henry Fayol (c) Robert Owen (d) None of these
- V. Which of the following could be a weakness?
(a) A developing market such as the Internet.
(b) Competitors with access to better channels of distribution.
(c) Poor quality of goods and services.
(d) Special marketing expertise
- VI. Which one of the following is NOT a determinant of Supply?
(a) Input Prices (c) Demonstration Effect
(b) Policy of Taxation (d) Expectations about future prices
- VII. The process of familiarizing the new employees to the organisation rules and regulations is known as _____.
(a) Placement (b) Induction (c) Recruitment (d) Selection
- VIII. The competitive advantage in the production function can be achieved through _____.
(a) high wastages (b) more cost (c) higher quality (d) lowest quantity
- IX. _____ is about supplying customers with what they want when they want it.
(a) JUT (b) JET (c) JAT (d) JIT
- X. The ultimate purpose of financial management is _____.
(a) to get a maximum return (c) to increase the wealth of owners
(b) to have a maximum risk factor (d) to get a maximum profit

GROUP: B (Answer any five questions)

(5 x 5 = 25)

2. Explain the concepts of Globalization, Liberalization and Privatization.
3. Describe the various kinds of organizational communication.

4. Elaborate the importance of economics in business world.
5. Explain the functions of management.
6. What is elasticity of demand? Suppose the price elasticity of demand for a good is -0.2. If there is a 5% increase in the price of the good, by what percentage will the demand for the good go down?
7. Explain the Garvin's 8 dimensions of quality.
8. What do you mean by the term industrial relations? What are its objectives?

(15 x 3 = 45)

GROUP: C (Answer any three questions)

9. (a) What is marketing mix? Describe 7P's of service marketing. (2+8)
- (b) Describe SWOT analysis with the help of an example. (5)
10. What do you mean by law of supply? What are the factors that affect supply? What are the exceptions to law of supply? (1+7+7)
11. Explain the concept of break even analysis. What are its advantages and disadvantages? (3+6+6)
12. What are the functions of material management? What are the motives of holding inventories? (10+5)

(3*5=15)

13. Differentiate between the following terms:

- i. Micro Economics and Macro Economics
- ii. Maslow's need hierarchy theory and Alderfer's ERG theory
- iii. Projection and Stereotyping
- iv. Change in demand and Change in quantity demanded
- v. Formal communication and Informal communication

Aliah University

Odd (Autumn)-Semester Examination - 2024

(For 7th Semester BTech(CSE) Programme)

Paper Name: Internet of Things

Paper Code: CSEUGPE12

Full Marks: 80

Time: 3 Hrs

Group - A

(5X2=10)

1. Give a one-sentence answer -

Remember, CO1-CO5

- a) What is a communication protocol widely used in IoT devices for constrained networks?
- b) What functions do sensors and actuators perform, respectively?
- c) What type of computing is required in an IoT system for real-time data processing closer to the devices?
- d) Which programming language is primarily used with Arduino?
- e) What is the primary operating system used by Raspberry Pi?

Group - B

(Answer any 6 questions)

(6X5=30)

- 2. Explain the key components of an IoT system with suitable examples. **Understand, CO1**
- 3. Differentiate between Sensing and Actuation. Provide real-life use cases for both. **Understand, CO1**
- 4. What is Machine-to-Machine (M2M) communication? Discuss its role in IoT. 2+3 **Understand, CO1**
- 5. Write a short note on any two communication protocols used in IoT. **Understand, CO1**
- 6. Describe the basic steps to integrate a temperature sensor with Arduino. Include the circuit setup and code snippet. **Apply, CO2**
- 7. Explain the concept of Interoperability in IoT with an example. Why is it important? 2+3 **Understand, CO3**
- 8. Discuss the role of Python programming in implementing IoT solutions. **Understand, CO3**

Group - C

(Answer any 4 questions)

(10X4=40)

9. a) Explain the architecture of an IoT-based smart home system. **5+5 Understand, CO4**
b) Discuss how fog computing enhances IoT applications. **5+5 Apply, CO2**
10. Write an Arduino code for a smart home application which will constitute of a Heater, Fan and Air Conditioner. Describe the code in details. **5+5 Apply, CO2**
11. Discuss the applications of IoT in the following fields:
a) Healthcare **5+5 Understand, CO5**
b) Agriculture **5+5 Understand, CO5**
12. Write a detailed note on the role of IoT in Smart Cities. Discuss any two use cases. **5+5 Understand, CO5**
13. a) Compare Cloud Computing and Fog Computing in the context of IoT. **5+5 Understand, CO4**
b) Explain the concept of the smart grid and its implementation in IoT. **5+5 Understand, CO4**
14. Propose an IoT solution for monitoring physical activity in elderly individuals. Include system architecture, components, and potential benefits. **5+5 Apply, CO5**

Aliah University
AUTUMN SEMESTER EXAMINATION 2024
FOR BTECH 4TH YEAR / 7TH SEM ONLY

Paper Name: Professional Elective III (Computer Vision)
Paper Code: CSEUGPE15

Full Marks: 80
Time: 3:00 Hrs

Section A
(Answer all questions)

5 × 2

1. (a) What is the size of a monochromatic image of dimension 1024x256 pixels? [CO1, Applying]
- (b) If red, green and blue intensities are represented with 1 byte each, what is the total number of colors? [CO2, Applying]
- (c) Write names of five applications of computer vision. [CO3, Understanding]
- (d) Mention a possible means for night vision. [CO3, Analyzing]
- (e) Define window and viewport. [CO4, Understanding]

Section B
(Answer any 6 questions)

6 × 5

2. Write the difference between spatial resolution and intensity resolution. [CO1, Understanding + Remembering] 5
3. How do illumination and reflectance contribute in formation of an image? [CO2, Remembering] 5
4. How does human vision work? Give a comparative view of cones and rods cells. [CO1, Analyzing] 3+2
5. Briefly mention the present-day challenges in computer vision. [CO3, Understanding] 5
6. Define image subtraction. Briefly outline how it is useful in X-ray videography. [CO2, Understanding + Remembering] 2+3
7. What is salt and pepper noise in images? Which filter is recommended for eliminating this type of noise? [CO4, Understanding + Analyzing] 3+2
8. How is a hyperplane represented? In object recognition, how are they useful? [CO5, Understanding + Analyzing] 2+3
9. What is the training and prediction time complexity of Nearest Neighbor classifier? [CO5, Analyzing] 5

Section C
(Answer any 4 questions)

4 × 10

10. With suitable diagrams, explain different image acquisition techniques. [CO1, Understanding + Remembering] 10
11. (a) Define image segmentation. [CO2, Understanding + Remembering] 3+7
(b) With mathematical expressions, derive how image averaging removes noises. [CO3, Understanding + Applying]
12. (a) Briefly explain spatial correlation and spatial convolution. [CO4, Understanding] 6+4
(b) What is histogram equalization? How is it useful? [CO2, Understanding]
13. (a) With a diagram, explain the 3D viewing process. [CO2, Understanding] 6+4
(b) Give a comparative view of parallel projection and perspective projection. [CO2, Analyzing]
14. (a) Briefly mention the steps in video processing of a vision application. [CO3, Understanding + Remembering] 6+4
(b) What is cross-validation in a classification problem? [CO5, Applying]
15. Write short notes. 5+5
(a) Edge detection [CO2, Remembering]
(b) Convolution and feature map [CO4, Understanding]